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A linear programming approach to the optimization of residential energy systems

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ABSTRACT

A decision-support tool (ORES) in the form of a linear program is developed to determine the optimal investment and operating decisions for residential energy systems. It shows how energy conversion units such as a cogeneration fuel cell, a heat pump, a boiler, photovoltaic panels and solar thermal collectors can be combined with energy storage devices, consisting of a battery and a hot water tank, to drive down total yearly energy costs and CO_2 emissions while meeting space heat, hot water and electricity needs. Under the assumption of perfect demand and production forecasts and depending on how the dwelling is allowed to exchange electricity with the grid, cost reductions between 5 and 60% are possible, whereas emissions can be cut by 45–90% with respect to a base case. Stochastic programming is used effectively to reduce the sensitivity to uncertainty in weather parameters. The resulting cost increase is limited to 1.2%. Decision rules are implemented to account for unforeseen variations in electric load. If it is assumed that peak loads can occur at any instant of the optimization horizon, energy costs rise by 9%, which in off-grid scenarios, are driven by the installation of an about 50% bigger battery system.

1. Introduction

Does it make economic or environmental sense today to power homes with locally converted energy? According to the Swiss Federal Office of Energy [1] residential demand is responsible for one third of the final energy consumption in Switzerland. In the European Union, the residential sector has – at 27% [2] – a slightly smaller share of the final consumption. Households use three types of energy: space heat, domestic hot water and electricity. It is noteworthy, that the heating load is about five times bigger than the hot water and the electric load. This article addresses the role of local conversion by proposing a flexible optimization set-up that considers the economic and technical parameters of decentralized energy conversion and storage systems, as well as the cost and carbon intensity of using grid electricity and natural gas.

There are many ways to provide for the energy needs of a building. One of the more innovative ones is to use a cogeneration fuel cell to produce both heat and electricity. With the increasing penetration of intermittent energy sources such as wind and solar, storage becomes more important to balance energy demand and supply – especially when exchanges with the electrical grid are limited. The present work models five residential energy conversion systems: a cogeneration fuel cell, a natural gas boiler, a heat pump, photovoltaic (PV) panels and solar thermal collectors. Furthermore, a battery and a heat tank are considered for energy storage. The question at hand is how to size and operate these systems in order to minimize annualized energy costs or CO_2 emissions or both. If storage was disregarded and if each unit could produce only either heat or electricity, the screening curve method as described for example by Buehring et al. [3] or more recently by Stoft [4] can give insights on the most economic investment. Since these conditions are not given in this case, linear programming is used to determine the investment and control decisions for a given set of weather, energy consumption and cost data. The advantage of linear programming is that it guarantees the most economic or least emission solution. It does however require a linear model of all conversion and storage units. The relative importance of minimizing CO_2 emissions over energy costs is set by attributing a cost to each ton of CO_2 emitted to the atmosphere. There are four stakeholders who may profit from the proposed tool:

1. households are informed about which energy system would best suit their needs,

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2. policy-makers can test the impact of subsidy schemes, such as feed-in-tariffs or net-metering, and steering taxes (e.g. a tax on CO_2 as it already exists in industry),
3. companies gain information about how to size their products to make them the most attractive to the residential market and
4. researchers can study how a new technology can be integrated into the residential energy mix.

Several scholars have worked on choosing the optimal set of energy conversion and storage units for buildings. Araz Ashouri has written his doctoral thesis on the simultaneous optimal design and control of energy systems on a building level [5]. He used a typical meteorological year (TMY) as input data for the outside temperature and the solar irradiation. His thesis does include combined heat and power systems, but he does not explicitly consider fuel cells. Ashouri's work is focused on applications to commercial and large office buildings, which due to their size tend to have a less fluctuating consumption than residential dwellings. The resulting optimization problem has a horizon of one year with hourly resolution. The choice of the time step is essential since it determines the demand and supply fluctuations that are covered by the optimization problem. Wright and Firth [6] have pointed out that averaging electric load data over 1 h periods significantly underestimates the load variability. The same is likely to hold true for hot water consumption and PV production, whereas space heat requirements may be less variable due to the thermal mass of the building. For the sake of computational tractability, this article follows Ashouri and the microgrid optimization packages presented below in choosing an hourly time step.

The U.S. Department of Energy (DOE) has been involved in the development of two software packages for the design of hybrid energy systems. The Lawrence Berkeley National Laboratory (LBNL) in cooperation with the California Energy Commission began to work on the Distributed Energy Resources Customer Adoption (DER-CAM) Model in 2000 [7]. The tool has been used mostly for the design of rather large microgrids [8–10]. Indeed, the smallest fuel cell that can be considered in the web based version of DER-CAM is 100 kW_{el} – about 40 times bigger than common peak loads in residential systems.

The microgrid modeling company HOMER Energy LLC was founded in 2009 with the goal of commercializing the Hybrid Optimization of Multiple Energy Resources Model (HOMER) developed at the National Renewable Energy Laboratory (NREL), a division of the U.S. DOE [11]. HOMER models a variety of energy technologies; among others, it includes a photovoltaic, a fuel cell, a boiler, and a battery system. It does not model a heat pump, a heat storage and solar thermal collectors. The program has been repeatedly used for the design of residential energy systems [12–15]. In these works, the focus usually lies on covering electric load. This is admissible, if it is presupposed that hot water and space heat loads are covered by either direct electric heating or by a heat pump, since it is ultimately electricity that provides the necessary heat. In this study, the choice of which technologies are responsible for covering heat loads is up to the optimization problem and not taken a priori. Another main difference is that the tool developed here, the Optimization of Residential Energy Systems (ORES), is based on linear programming, meaning that it solves an optimization problem specified by a cost function and a set of constraints, whereas HOMER uses decision rules to explore user-specified design points for the sizes of the energy systems to be installed. While this work was under review HOMER released a derivative free optimization approach inspired by genetic algorithms that mostly eliminates the need to specify discrete design points. It is a non-deterministic approach meaning that HOMER is not guaranteed to find the optimal investment and dispatch decisions. However, due to its simulation based nature, HOMER can capture non-linearities in the energy system. Furthermore, since the HOMER dispatch is governed by decision rules based on experience, it may be more robust towards uncertain demand and supply forecasts than the deterministic version of ORES. Under two admittedly strong assumptions, ORES is guaranteed to find the optimal investment and dispatch decisions. The first is that all cost and conversion parameters are independent from the investment and control decisions. This is a prerequisite for the model to be linear. The second assumption is that all parameters are known deterministically for the entire planning horizon, i.e. for one year, with hourly resolution.

Sections 2 and 3 explain how the household energy systems are modeled. Subsequently, the performance of ORES is benchmarked against HOMER. A comparison between the ORES designed system with a conventional system consisting of a gas boiler for heat and access to the electrical grid will show that – for the Swiss case – the integration of alternative energies can reduce both yearly energy costs and carbon emissions. An analysis of 12 years of meteorological data for the town Ecublens, Switzerland, shows that stochastic programming can be used efficiently to render the program more robust towards changes in weather parameters. Finally, the impact of mitigating load uncertainty through decision rules is investigated.

2. Methodology

The energy flow diagram (Fig. 1) shows from left to right the energy sources, conversion units, storage units and the energy utilization that may be part of a residential energy system. The aim of the optimization problem is to route energy from left to right, i.e. from source to utilization, with the lowest cost. It is possible to optimize only a subset of conversion and storage units such as for example a PV system, a fuel cell and a battery.

For each time-step $t \in T$, where T is the horizon of the optimization problem, each conversion unit relates at least one element s of the set of energy sources

$$S = \{\text{Solar}(sun), \text{Electricity}(el), \text{Gas}(gas)\}$$

to at least one element u of the set of energy services.

$$U = \{\text{Space Heat(sph)}, \text{Domestic Hot Water(dhw)}, \text{Electricity(el)}\}$$

Each conversion unit is described by three parameters (all in W): the installed capacity x, the inflowing power $x^+_{s,t}$ and the outflowing power $x^-_{u,t}$ at each time-step. Similarly, storage units are described by the installed capacity y in J, the inflowing power $y^+_{u,t}$ and the outflowing power $y^-_{u,t}$. The power flowing out of a storage unit must be of the same type as the power flowing into a storage unit. The optimization problem based on the energy flow diagram takes the form:

$$\underset{x,x^+,x^-,y,y^+,y^-}{\text{minimize}} \quad \boldsymbol{c}_x^{\mathrm{T}}\boldsymbol{x} + \boldsymbol{c}_y^{\mathrm{T}}\boldsymbol{y} + \Delta t \sum_t \boldsymbol{c}_{op}\boldsymbol{x}^+{}_t \tag{2.1a}$$

$$\text{subject to} \quad {}_{\boldsymbol{K}}\boldsymbol{x} \leq \boldsymbol{x}_t^+ \leq \boldsymbol{K}\boldsymbol{x} \qquad \forall t \tag{2.1b}$$

$$\boldsymbol{0} \leq \boldsymbol{H}^-\boldsymbol{x}_t^- \leq \boldsymbol{H}^+\boldsymbol{x}_t^+ \qquad \forall t \tag{2.1c}$$

$$\boldsymbol{A}\boldsymbol{x}_t^- - \boldsymbol{B}\boldsymbol{y}_t^{\pm} = \boldsymbol{L}_t \qquad \forall t \tag{2.1d}$$

$$\boldsymbol{0} \leq \boldsymbol{y}_t^+ \leq \boldsymbol{P}^+\boldsymbol{y} \qquad \forall t \tag{2.1e}$$

$$\boldsymbol{0} \leq \boldsymbol{y}_t^- \leq \boldsymbol{P}^-\boldsymbol{y} \qquad \forall t \tag{2.1f}$$

$$\begin{aligned}\boldsymbol{M}_0\boldsymbol{y} \leq \Big(&\sum_{\tau=1}^{t}\boldsymbol{E}^{t-\tau}(\boldsymbol{H}^+\boldsymbol{y}_\tau^+ - \boldsymbol{H}^-\boldsymbol{y}_\tau^-) \\ &-(\boldsymbol{M}_1 + \boldsymbol{M}_2\boldsymbol{y})\Big)\Delta t + \boldsymbol{E}^t\boldsymbol{y}_0 \leq \boldsymbol{y} \qquad \forall t\end{aligned} \tag{2.1g}$$

$$\Big(\sum_{\tau=1}^{T}\boldsymbol{E}^{T-\tau}(\boldsymbol{H}^+\boldsymbol{y}_\tau^+ - \boldsymbol{H}^-\boldsymbol{y}_\tau^-) - (\boldsymbol{M}_1 + \boldsymbol{M}_2\boldsymbol{y})\Big)\Delta t + \boldsymbol{E}^T\boldsymbol{y}_0 = \boldsymbol{y}_T \tag{2.1h}$$

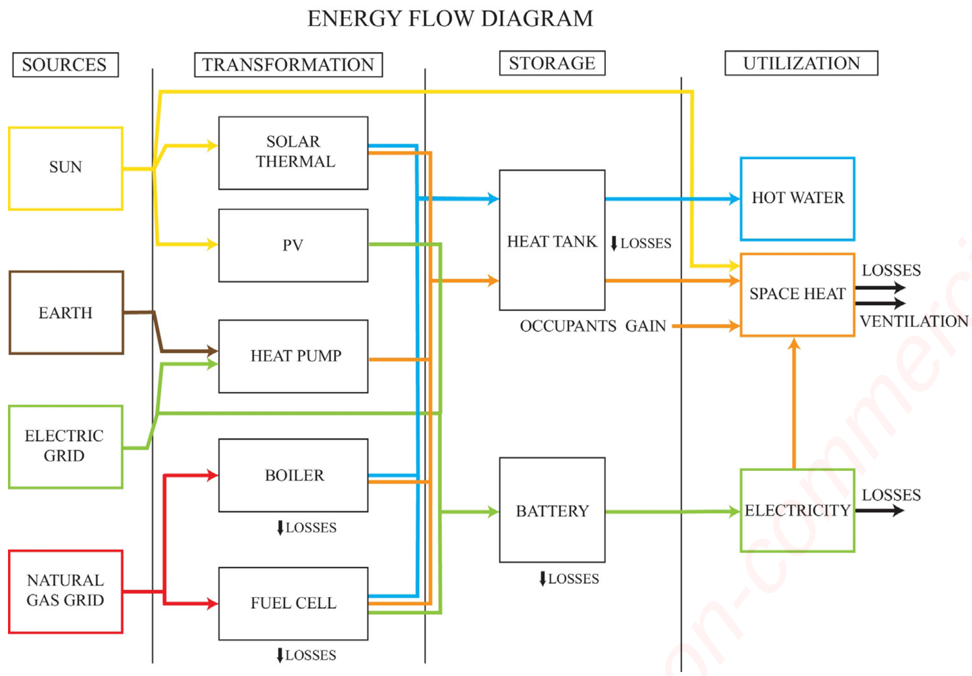


Fig. 1. Energy flow diagram.

where $\boldsymbol{x}, \boldsymbol{x}_t^+, \boldsymbol{x}_t^-, \boldsymbol{y}, \boldsymbol{y}_t^+, \boldsymbol{y}_t^-$ are vectors that contain the decision variables $x, x_{s,t}^+, x_{u,t}^-, y, y_{u,t}^+, y_{u,t}^-$ of each technology. The difference between the power flowing into and out of storage is denoted as $\boldsymbol{y}_t^\pm = \boldsymbol{y}_t^+ - \boldsymbol{y}_t^-$. Table 1 explains the remaining problem parameters.

A typical time horizon that includes seasonal trends is one year. If a hourly resolution is chosen, this means that each decision dependent on t corresponds to 8760 variables. A linear model was chosen because it can be solved efficiently by available solvers. When all conversion and storage units are modeled, the problem consists of about 193,000 decision variables and 385,000 constraints. It is important to solve the deterministic problem efficiently, since a stochastic programming approach is presented to account for uncertainty in weather parameters. Stochastic programming increases the computational complexity in proportion to the number of considered weather scenarios as explained in Section 3.5.

The following four sections describe the modeling of the units in the energy flow diagram. For the energy sources and utilization it was necessary to find time-series data that describe the solar radiation, ambient temperature, the electricity and the hot water consumption. The modeling of the energy conversion and storage units requires to examine their underlying physical principles and their costs. These parameters are summarized in Tables 4 and 5, whereas the decision variables used to model the units are shown in Table 6. All data is given for an average three-person household located in Ecublens, Switzerland.

Table 1
Problem parameters.

Objective function	
$\boldsymbol{c}_x$, $\boldsymbol{c}_y$	are vectors consisting of the annualized investment and maintenance costs (CHF W^{-1} yr^{-1}) of the conversion and storage units respectively. The investment cost P is annualized over the unit lifetime n in years while considering a yearly interest rate r according to:$A = P\frac{r(1+r)^n}{(1+r)^n-1}$The interest rate is assumed to be 3%.
Δt	is the length of a time-step (s).
$\boldsymbol{c}_{op}$	expresses the cost of using the conversion units (CHF J^{-1}). It consists of the fuel cost, the maintenance cost and the cost of emitting CO_2. The latter is calculated based on the carbon intensity of the fuel.
Conversion units	
$\boldsymbol{\kappa}$, $\boldsymbol{K}$-	are diagonal matrices specifying the minimum respectively maximum operating thresholds of the conversion units.
$\boldsymbol{H}^+$	is a diagonal matrix that contains the efficiencies for converting primary energy into intermediary energy types, such as heat, or into energy services.
$\boldsymbol{H}^-$	is the identity matrix except for the cases in which intermediary energy types are allocated to energy services.
Energy demand	
$\boldsymbol{A}$, $\boldsymbol{B}$	are incidence matrices that relate conversion respectively storage units to energy services.
$\boldsymbol{L}_t$	is the demand of energy services (W) at time t.
Storage units	
$\boldsymbol{P}^+$, $\boldsymbol{P}^-$	are diagonal matrices describing the charge respectively discharge power density (W J^{-1}) of the storage units.
$\boldsymbol{M}_0$	is a diagonal matrix expressing the maximum depth of discharge of the storage units.
$\boldsymbol{E}$	is a diagonal matrix detailing the self-discharge of the storage units.
$\boldsymbol{H}^+$, $\boldsymbol{H}^-$	are diagonal matrices describing the charge respectively discharge efficiency of the storage units.
$\boldsymbol{M}_1$, $\boldsymbol{M}_2$	account for storage losses that are independent of the current state of charge.
$\boldsymbol{y}_0$, $\boldsymbol{y}_T$	express the initial respectively terminal state of charge of the storage units.

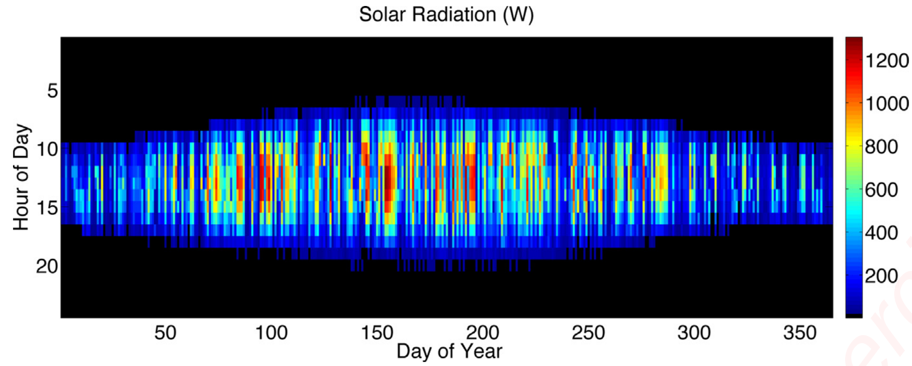


Fig. 2. Solar radiation on a 30° inclined, south-facing roof: 1265 kWh per year.

3. Model

3.1. Energy sources

Whereas the natural gas and the electricity grid represent dispatchable sources of energy, solar energy exhibits strong intermittency. It undergoes both daily and seasonal cycles (Fig. 2) The radiation data is based on ground-measurements from Meteotest [16]. The Swiss electricity market is expected to be liberalized in 2018 [17], whereas for the Swiss gas market no liberalization is planned [18]. As it stands households have to choose the energy supplier responsible for their region. Romande Energie supplies Ecublens with electricity for 22.20 cts per kWh (in the flat tariff regime) while charging a yearly connection fee of 85 CHF [19]. The gas supply is assured by the Services Industriels de Lausanne which charge 11.86 cts per kWh_{HHV} and a yearly connection fee of 219 CHF [20]. At a depth of several meters, the earth is assumed to be at almost the same temperature throughout the year, which allows the ground-source heat pump to have a constant coefficient of performance [21].

3.2. Conversion technologies

Each conversion technology together with its characteristic parameters is described below. Furthermore, the constraints that govern the use of the technology and the impact of its use on the objective function are detailed. For all technologies it is assumed that there is an electricity and a heat sink, that allows for the evacuation of any excess electricity or heat that may result from the system operation.

3.2.1. Cogeneration fuel cell

A cogeneration fuel cell allows to convert the chemical energy of natural or biogas into both heat and electricity. This work focuses on solid oxide fuel cells (SOFC). Due to their high internal temperature of between 700 and 900 °C, SOFCs are less sensitive to gas impurities, have better reaction kinetics and thus a higher electrical conversion efficiency than fuel cells that work at lower temperatures [22,23]. In order to guarantee the longevity of the fuel cell stack, the number of thermal cycles on the system should be limited. Indeed, in the Japanese Ene-Farm project, the SOFCs run continuously, whereas lower temperature fuel cells such as proton exchange membrane fuel cells can be shut down at night, when energy demand is low [24]. This study examines the performance of a 2.5 kW_{el} commercially available SOFC system that produces hot water at about 70 °C [25]. It is assumed that domestic hot water is required – and if applicable stored – at a temperature of 60 °C. This is to avoid the risk of legionella infections [26]. Space heating is assumed to require heat at a temperature of 40 °C. For modern floor heating techniques, even 30 °C can be sufficient. The fuel cell unit can thus produce both space heat and domestic hot water. Its electric and the thermal efficiency are based on the lower heating value of natural gas. The estimated system lifetime of fifteen years accounts for stack replacements every five years.

Concerning the constraints governing the operation of the SOFC (Eq. (3.1)), it is assumed that the gas flow rate can be modulated between 30 and 100% of the rated capacity. It is furthermore assumed that the electrical and thermal efficiency of the fuel cell stay constant over the entire modulation range. For the electrical conversion, this introduces an error of about 20%, whereas the thermal efficiency is almost two times higher at full load than at a load factor of 30% (Fig. 3). Restraining the fuel cell operation range or piecewise linearization can be used to obtain a more realistic simulation. The first approach takes away some of the flexibility of the fuel cell and may thus lead to suboptimal results. The second approach requires the introduction of binary variables, which increases the computational complexity of the optimization problem. Ramping the output power through the entire modulation range takes about 15 min. Since the time-step of the simulation is 1 h, no ramping constraints need to be introduced in the model. The assumption of a heat and electricity sink allows the thermal and electrical output to vary between zero and the output based on the maximum gas flow rate, since any excess production that may result from the fuel cell operation can be evacuated.

$$\forall t: \quad \begin{aligned} & \underline{k} \cdot x_{\text{gas}} \leq x^{+}_{\text{gas},t} \leq x_{\text{gas}} \\ & 0 \leq x^{-}_{\text{el},t} \leq \eta_{\text{el}} \cdot x^{+}_{\text{gas},t} \\ & x^{-}_{\text{sph},t} + x^{-}_{\text{dhw},t} \leq \eta_{\text{th}} \cdot x^{+}_{\text{gas},t} \\ & 0 \leq x^{-}_{\text{sph},t}, x^{-}_{\text{dhw},t} \end{aligned} \tag{3.1}$$

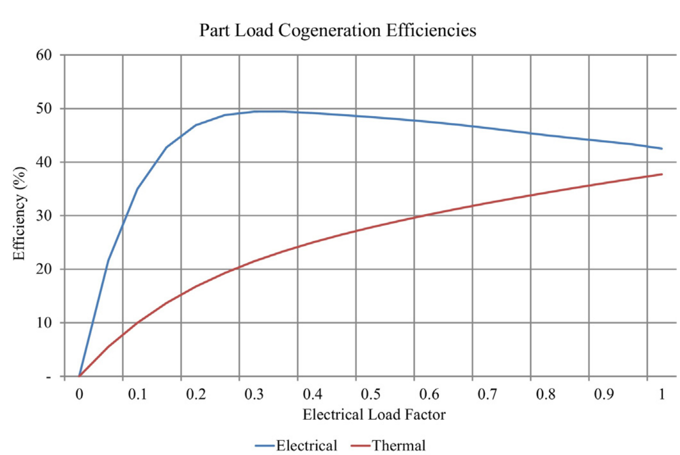


Fig. 3. From a load factor of 0.3 to full load, the electrical efficiency drops by 20%, whereas the thermal efficiency doubles [27].

The cost of operating the fuel cell consists of the capital investment, maintenance and fuel expenses as well as the cost of carbon:

$$c_{\mathrm{cap}} \cdot \eta_{\mathrm{el}} \cdot x_{\mathrm{gas}} + \Delta t \sum_{t} \left(c_{\mathrm{fuel}} + c_{\mathrm{CO_2}} \rho_{\mathrm{CO_2}} \right) x^{+}_{\mathrm{gas},t} \tag{3.2}$$

3.2.2. Photovoltaic modules

Photovoltaic modules transform solar radiation (I) into electricity. The world market is largely dominated by silicon based modules, although other technologies such as thin-film cells are on the rise. There are three main types of silicon based PV cells: mono-crystalline, poly-crystalline and amorphous cells. All of these technologies have similar retail prices. It is assumed that the performance of the PV system is independent on the outside temperature and that the efficiency of the inverter used to convert the produced direct current into alternative current does not depend on the power output of the PV plant. At 2500 CHF per kW the installation cost may seem to be quite high. Indeed, a recent report by the Fraunhofer Institute for Solar Energy Systems (ISE) [28] shows that retail prices in Germany stand at around 1600 Euro per kW. In Switzerland the PV industry is not as developed as in Germany. According to the REsource platform of the International Renewable Energy Agency (IRENA) [29], the installed capacity in Switzerland was 1 GW versus 38 GW in Germany for the year 2014.

The main constraint governing the conversion of solar energy into electricity is the direct dependence on the incoming radiation. If the radiation is higher than at the standard rating conditions (1000 $\mathrm{W m^{-2}}$), the system may produce more power than its nominal output.

$$\forall t: \quad 0 \le x^{-}_{\mathrm{el},t} \le \tilde{x}_{\mathrm{el},t} \tag{3.3}$$

where

$$\tilde{x}_{\mathrm{el},t} = \frac{\eta_{\mathrm{el}} \cdot \eta_{\mathrm{pr}} \cdot I_t}{1000\,\mathrm{W m^{-2}}} \cdot \frac{x_{\mathrm{el}}}{\rho}$$

is the maximum amount of PV power that can be produced at a given time-step. The cost of operating the photovoltaic system consists exclusively of the capital investment and the maintenance expenses: $c_{\mathrm{cap}} \cdot x_{\mathrm{el}}$.

3.2.3. Solar thermal collectors

Solar thermal collectors capture the direct and diffuse solar radiation, which heats up a fluid that circulates through the collectors. In general, solar thermal collectors can be specified by their net absorption efficiency (η) with respect to the incoming solar radiation and their conduction and convection losses (k) per square meter of collector area. The conduction and convection losses depend on the difference between the collector and the ambient temperature. The Solar Rating & Certification Corporation, an US rating agency for solar equipment makes its ratings publicly available [30]. The system installation cost is highly variable. The Swiss Energyscope, an online calculator for the future Swiss energy mix developed by EPFL, assumes values between 500 and 5500 CHF per $\mathrm{kW_{th}}$ [31].

The heat produced by the solar thermal system can be used for both space heating and domestic hot water purposes (Eq. (3.4)). The exact split is an outcome of the optimization problem. The formulation of how this split depends on the collector temperature (T_{coll}) has been omitted. Due to the heat losses of the collector, the produced thermal energy depends on the difference between the collector and the ambient temperature (T_{amb}). The surface area of the solar thermal system is the amount of thermal capacity to be installed (x_{th}) divided by the typical power density (ρ) that can be achieved by solar thermal collectors in Switzerland.

$$\forall t: \quad \begin{array}{l} x^{-}_{\mathrm{sph},t} + x^{-}_{\mathrm{dhw},t} \le \left(\eta \cdot I_t - k \cdot \left(T_{\mathrm{Coll},t} - T_{\mathrm{Amb},t} \right) \right) \cdot \dfrac{x_{\mathrm{th}}}{\rho} \\ 0 \le x^{-}_{\mathrm{sph},t}, x^{-}_{\mathrm{dhw},t} \end{array} \tag{3.4}$$

The cost of the electricity required to drive a pump for the solar thermal system is neglected. If judged necessary it can be modeled as a function of $x^{-}_{\mathrm{sph},t} + , x^{-}_{\mathrm{dhw},t}$. The cost of operating the system consists thus exclusively of the capital investment and the maintenance expenses: $c_{\mathrm{cap}} \cdot x_{\mathrm{th}}$.

3.2.4. Heat pump

A heat pump allows to reverse the natural flow of heat from warmer to cooler temperatures through the use of electricity. A compressor and an expansion valve are used to evaporate a gas at low temperatures and to condense it at higher temperatures, thus transferring heat from a colder to a warmer reservoir. The coefficient of performance (η_{cop}) of a heat pump measures the ratio between the electricity required by the compressor and the heat given to the warmer reservoir. It can be estimated through the Carnot efficiency of a heat machine working between a cold and a hot reservoir. In the present model, a shallow-soil heat pump is used for space heating, meaning that it needs to provide heat at about 45 °C (T_{hot}). The cold reservoir is the earth at a depth of several meters, where the temperature remains almost constant at about 11 °C (T_{cold}) throughout the year. Discounting the inverse Carnot efficiency with a factor of 0.45, this gives a η_{cop} of:

$$\eta_{\mathrm{cop}} = 0.45 \cdot \frac{T_{\mathrm{hot}}}{T_{\mathrm{hot}} - T_{\mathrm{cold}}} = 0.45 \cdot \frac{318.5\,\mathrm{K}}{34.0\,\mathrm{K}} \approx 4 \tag{3.5}$$

The space heat production of the heat pump is only constrained by the installed capacity.

$$\forall t: \quad 0 \le x^{-}_{\mathrm{sph},t} \le \eta_{\mathrm{cop}} \cdot x_{\mathrm{el}} \tag{3.6}$$

The cost of operating the heat pump consists of the capital investment, the maintenance expenses, the electricity used to power the heat pump and the cost of carbon:

$$c_{\mathrm{cap}} \cdot x_{\mathrm{el}} + \Delta t \sum_{t} \left(c_{\mathrm{fuel}} + c_{\mathrm{CO_2}} \rho_{\mathrm{CO_2}} \right) \frac{x^{-}_{\mathrm{sph},t}}{\eta_{\mathrm{cop}}} \tag{3.7}$$

3.2.5. Condensing gas boiler

Condensing boilers extract the higher heating value of the fuel they are burning, because they are able to recuperate the heat released when the water vapor, that is generated by any combustion of hydrocarbons, condenses. For natural gas, the difference between the lower and the higher heating value is about 10%. The conversion of the chemical energy stored in natural gas to heat is very efficient. Since inevitably, some of the produced heat will leave the building through the chimney, the boiler is assumed to have a 95% efficiency (η), which is slightly lower than the efficiency given by the Swiss Energyscope. It is assumed that the boiler can be used for catering to both space heat and domestic hot water needs. The split between the heat used for space heating and domestic hot water is subject to the optimization problem.

$$\forall t: \quad \begin{array}{l} x^{+}_{\mathrm{gas},t} \le x_{\mathrm{gas}} \\ 0 \le x^{-}_{\mathrm{sph},t} + x^{-}_{\mathrm{dhw},t} \le \eta \cdot x^{+}_{\mathrm{gas},t} \\ 0 \le x^{-}_{\mathrm{sph},t}, x^{-}_{\mathrm{dhw},t} \end{array} \tag{3.8}$$

The cost of operating the boiler consists of the capital investment, the maintenance expenses, the gas and the cost of carbon:

$$c_{\mathrm{cap}} \cdot \eta \cdot x_{\mathrm{gas}} + \Delta t \sum_{t} \left(c_{\mathrm{fuel}} + c_{\mathrm{CO_2}} \rho_{\mathrm{CO_2}} \right) x^{+}_{\mathrm{gas},t} \tag{3.9}$$

3.3. Storage technologies

Since some of the energy production and most of the consumption is intermittent in nature, storage becomes important to balance energy demand and supply. Warm water tanks have been in use for a long time, whereas the use of batteries is mostly restricted to regions with no or non reliable access to the electric grid. In general, storage technologies are characterized by their cost, their round-trip efficiency and their power versus energy density. In some cases, it may be necessary to restrict the size of the storage devices, in order to ensure that they still fit into the building.

3.3.1. Battery

There are two units that can produce electricity in the system under consideration: the fuel cell and the PV modules. Both of them produce DC power. The DC/AC converters, which are necessary to use this DC power to serve the household AC loads or to export power to the grid, are included in the investment cost of the respective unit. It is assumed that a battery can be installed in the household energy system without incurring further costs for the conversion chain. In the case where there is no power production, the battery could still be used to take advantage of variations in electricity prices. There is a variety of different battery technologies on the market. IRENA has recently published a review about the available technologies [32]. Lead-acid based batteries continue to have the lowest investment cost, however they usually have a lower depth of discharge and a lower cycle life than lithium-ion based batteries. A case study of the German battery market found that a lithium-ion battery had the lowest investment cost per kWh of electric energy throughput. Another advantage is that lithium-ion batteries have higher energy and power densities than lead acid batteries. This makes it easier to install them in residential buildings, since they require less space. For the Ecublens case study, the installation of a commercially available battery system is considered [33]. The lifetime of the battery is calculated based on the assumption that on average there is one charge-discharge cycle per day.

The utilization of the battery is governed by the capacity constraint, the maximum depth of discharge, the round-trip efficiency, the self-discharge, and the maximum amount of charging and discharging power that the battery can handle. The initial and terminal conditions determine the state of charge at the beginning and at the end of the optimization horizon. They are used to avoid end-of-horizon effects. An additional constraint ensures that the battery is not cycled more than once per day on average – a condition that aims at increasing the battery lifetime (Eq. (3.10)).

$$
\begin{aligned}
\forall t: \quad & m_0 y \le \Delta t \sum_{\tau=1}^{t} \epsilon^{t-\tau}\left(\eta^+ y^+_{\mathrm{el},\tau} - \eta^- y^-_{\mathrm{el},\tau}\right) + \epsilon^t y_0 \le y \\
& \Delta t \sum_{\tau=1}^{T} \epsilon^{T-\tau}\left(\eta^+ y^+_{\mathrm{el},\tau} - \eta^- y^-_{\mathrm{el},\tau}\right) + \epsilon^T y_0 = y_T \\
& 0 \le y^+_{\mathrm{el},t} \le \rho^+ y \\
& 0 \le y^-_{\mathrm{el},t} \le \rho^- y \\
& \sum_{\tau=1}^{T} y^-_{\mathrm{el},\tau} \le \frac{T}{24} \cdot \eta^- \cdot (1-m_0) \cdot y
\end{aligned}
\tag{3.10}
$$

The charging and the discharging efficiencies are the geometric mean of the roundtrip efficiency (η): $\eta^+ = \sqrt{\eta}$; $\eta^- = 1/\sqrt{\eta}$. The cost of using the battery consists solely of the capital investment: $c_{\mathrm{cap}} \cdot y$.

3.3.2. Heat store

A challenge in representing the warm water tank lies in finding a linear model. The capacity of the tank depends on its volume, whereas the associated heat losses depend on the tank surface. Both the surface and the volume depend on the tank geometry. In order to create a linear dependence between the two, it is assumed that the tank is a cylinder with a fixed diameter. Typical diameters found on the Swiss market range from 0.5 to 5 m [34]. The tank diameter is a model parameter and can be easily adjusted to correspond to the system under examination. The height of the cylinder is subject to optimization. If needed, the height can be constrained to ensure a smooth building integration of the store. Since the density of water declines with rising temperature, there will be a temperature stratification in the tank. The influence of the heat losses on the tank temperature is not modeled. It is assumed that the upper part of the domestic hot water reservoir is at a constant temperature of 60 °C (T_{hot}), whereas the inflowing water is at a temperature of 10 °C (T_{cold}) [35]. The heat store capacity and losses are calculated based on its height (y_h):

$$
\begin{aligned}
& y = \rho c_p (T_{\mathrm{hot}} - T_{\mathrm{cold}}) \pi \left(\frac{\phi}{2}\right)^2 \cdot y_h \\
& m_1 = \pi U (T_{\mathrm{hot}} - T_{bdg}) \frac{\phi^2}{2} \\
& m_2 = \pi U \left(\frac{T_{\mathrm{hot}} - T_{\mathrm{cold}}}{2} - T_{bdg}\right) \phi y_h
\end{aligned}
\tag{3.11}
$$

The heat losses to the environment depend on the difference between the storage temperature and the building temperature. It is not possible to use the exact building temperature because this would render the model non-linear. Instead, the mean between the highest and lowest allowable building temperature is used to calculate the heat losses. The error on the temperature difference between the heat store and the building is below 10%. As for the battery, initial and terminal conditions define the initial and terminal state of charge of the tank (Eq. (3.12)). Since no distinction needs to be made between inflowing and outflowing energy, the decision variable $y^{\pm}$ describes the net flow into the store. It can take positive and negative values.

$$
\begin{aligned}
\forall t: \quad & 0 \le \Delta t \left(\sum_{\tau=1}^{t} y^{\pm}_{\mathrm{dhw},t} - m_1 - m_2 y\right) + y_0 \le y \\
& \Delta t \left(\sum_{\tau=1}^{T} y^{\pm}_{\mathrm{dhw},t} - m_1 - m_2 y\right) + y_0 = y_T
\end{aligned}
\tag{3.12}
$$

The cost of using the heat store consists solely of the capital investment: $c_{\mathrm{cap}} \cdot y$.

3.4. Energy utilization

Load duration curves (LDC) show the variability of energy demand [36]. Originally developed for the electric power industry, they can also be applied to space heat and domestic hot water (Fig. 4). To appreciate the demand variability, a distinction is made between primary, non-controllable, and secondary, controllable, load. The electric power consumption of the heat pump for example is subject to optimization. It is thus considered a secondary and not a primary load. Unsurprisingly, the space heat requirements are higher in winter than in summer. Since there is an inherent storage due to the thermal mass of the building, the space heat demand is less erratic than the domestic hot water demand. Indeed, the domestic hot water demand is characterized by infrequent high peak loads compared with low average consumption. This motivates the use of a hot water tank to smooth out the load variations and to enable a more steady hot water production, which allows the installation of smaller conversion units. Finally, one needs to keep in mind, that energy demand is modeled on a hourly time-basis. As Wright and Firth [6] point out averaging electricity consumption data over 1 h leads to

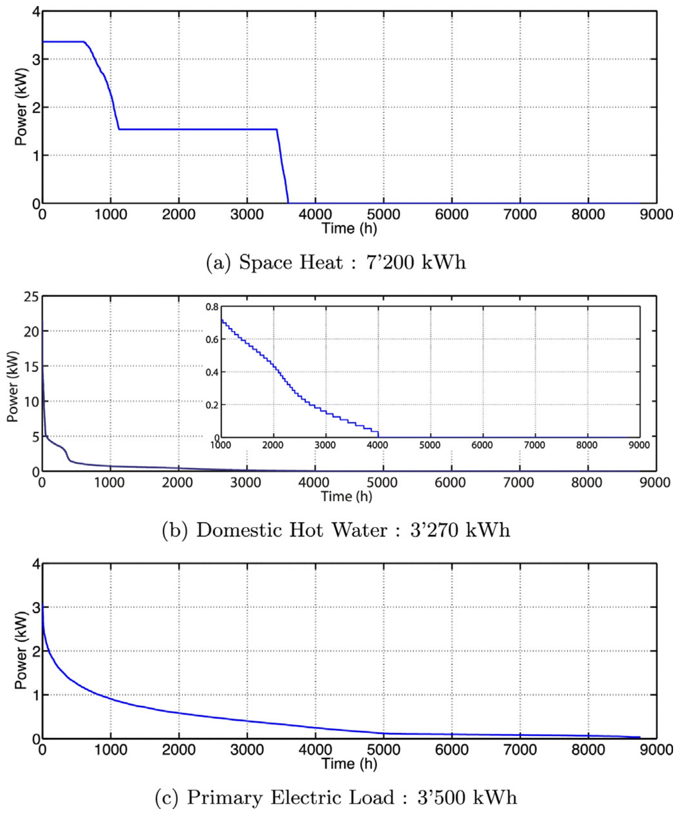


Fig. 4. Load duration curves with zoom-up on the hot water consumption after the first 1000 h.

an underestimation of the load variability. This may well hold for heat loads as well.

Residential hot water consumption data is not readily available with a high time resolution. Most utilities check the water, gas and electricity consumption of their customers on a yearly basis. A possible way of constructing the consumption data is to model the hot water appliances in a household and their likely usage [37]. The demand used in this study is based on the consumption profiles developed by Jordan and Vajen [38]. It is scaled to correspond to the average consumption of a Swiss three person household [1].

Richardson et al. created a spreadsheet model that simulates the electricity demand and the number of present active (i.e. non-sleeping) occupants of a UK household based on the number of occupants, the installed appliances, whether it is a weekday or a weekend, and on the month of the year [39]. Their model generates 24 h load profiles with a 1 min resolution. Their data has been aggregated to yield yearly load profiles with hourly resolution. No appliances that use electricity to generate space heat or domestic hot water were considered when establishing the load profiles. Finally, the resulting data was scaled to correspond to the average of a Swiss three person household.

The space heat requirements of the building are calculated using the 1R1C model [5,40] (Fig. 5), which represents the building by a single thermal zone. The current source represents the thermal gains $Q^+(t)$ within the building. These may consist of the heating, solar, and internal gains as well as the losses from the heat store. Internal gains represent the heat produced by the occupants of and the appliances in the building. The capacitance C models the heat storage effect associated with the thermal mass of the building. The transmittance U or conductivity represents the building insulation. It accounts for heat losses through the walls, windows, roof, base, and air changes.

Both C and U depend on the building floor area. For a three person household, an area equal to three times the Swiss per capita average of 56 m_2 [1] is assumed. The total thermal transmittance was calculated based on a rough estimate of 1 W/K per square meter of floor area. The total thermal capacitance is established by estimating the volume proportions of different materials in the building. It is assumed that 90% of the volume is made up by air, 5% by wood and 5% by brick. The effect of glass on the thermal mass is neglected. The possibility to increase natural convection through an opening of windows that goes beyond guaranteeing the amount of air changes that are necessary to avoid stale air in the building is modeled by the term $Q^-(t)$.

The discrete time equation yielding the building temperature at time-step t, $T_{bdg}(t)$, in dependence of the ambient temperature $T_{amb}(t)$ is implemented as a constraint in the linear program (eq. (3.13)). Since the temperature at time t depends on the temperature at $t-1$, the formulation of this constraint is more computationally intensive than the formulation of constraints that are independent of future or previous time-steps.

$$\forall t:$$

$$T_{bdg,t} = e^{\frac{-\Delta t}{\tau}} \cdot T_{bdg,t-1} + \left(1 - e^{\frac{-\Delta t}{\tau}}\right) \cdot T_{amb,t} + \frac{1}{U} \cdot \left(1 - e^{\frac{-\Delta t}{\tau}}\right) \cdot (Q_t^+ - Q_t^-) \tag{3.13}$$

where $\tau = C/U$ is the time constant associated with the building and T_{bdg}, T_{amb} are in Kelvin. The ambient temperature data is included in the weather files [16] that were used earlier for the calculation of the solar radiation.

The control of the space-heating system is subject to optimization. It is considered that the residents' goal is to maintain the room temperature in a comfort zone between T_{min} and T_{max}. The auxiliary variable ΔT measures the deviation from this comfort zone. Any deviation is punished by an user assigned penalty c_T. The problem of optimizing the space heat control is formulated using soft constraints (Eq. (3.14)),

$$\forall t: \quad \begin{aligned} &\Delta T_t \geq T_{bdg,t} - T_{max} \\ &\Delta T_t \geq T_{min} - T_{bdg,t} \\ &\Delta T_t \geq 0 \end{aligned} \tag{3.14}$$

while adding the following cost term to the objective function: $c_T \cdot \sum_t \Delta T_t$.

3.5. Stochastic programming and operating reserve

The optimization problem exhibits two kinds of uncertainty: parameter and time-series uncertainty. Parameter uncertainty pertains to the uncertainty in the cost and efficiency parameters of the conversion and storage units, as well as to fuel and electricity prices. It can be partly mitigated through careful supplier examination. In the worst case, if the cost parameters are not as expected, this will result in higher yearly energy costs. The system will however still be able to meet energy demand. This is why it seems more important to deal with time-series uncertainty, i.e. the

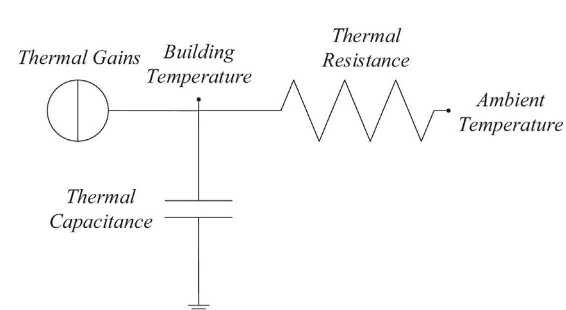


Fig. 5. The 1R1C model.

uncertainty on meteorological conditions and energy demand. If these characteristics are not estimated correctly, there is a risk that the system will not be able to meet energy demand – be it electricity, space heating or hot water – at all times. It is assumed that guaranteeing occupant comfort is paramount, which is why this uncertainty is investigated in more detail. The analysis focuses on meteorological data because radiation [41] and temperature [42] data can both be inferred from satellite measurements and are thus rather readily available, whereas it is more complicated to find multi-year measurements of residential electricity and hot water consumption. The analysis of weather data from 2005 to 2014 for Ecublens, Switzerland, has shown that a system sized for a typical meteorological year is *not* able to meet electricity and space heat demand in five out of ten years. A shortage of electric capacity is only relevant in off-grid settings, whereas the shortage in space heat is important in all settings. In order to find a solution for the investment decision that satisfies energy demand in all cases, the horizon of the optimization problem is extended to three years, consisting of a cold, a temperate and a warm year. There are two reasons for which no more than three years are considered in the stochastic program. The first is computational complexity. Optimizing over one year for the off-grid case takes about 200 s. Extending the time-horizon to three years takes about 2000 s – an increase of computational complexity by a factor 10. The second reason is that the aim lies in evaluating the stochastic solution by examining its out-of-sample performance, i.e. its performance on weather scenarios that were not part of the optimization procedure. Since no information about the respective likelihood of each weather scenario is available, it is assumed that they are all equally likely. They are thus assigned the same weight in the stochastic program. The investment decisions are considered as first-stage decisions, which means that they need to be the same for all three years. The operating decisions are considered as second stage decisions and can be adapted to each weather scenario. The number of second stage decision variables is proportional to the number of weather scenarios (eq. (3.15)) [43].

$$\underset{x,x^+,x^-,y,y^+,y^-}{\text{minimize}} \; \boldsymbol{c_x}^{\mathsf{T}}\boldsymbol{x} + \boldsymbol{c_y}^{\mathsf{T}}\boldsymbol{y} + \frac{1}{N}\sum_{n=1}^{N}\Delta t\sum_{t}\boldsymbol{c}_{op}\boldsymbol{x}^{+}{}_{t,n} \tag{3.15}$$

where N is the number of years considered in the stochastic program. The constraints are the same as in the deterministic problem, only that they now need to hold for each weather scenario.

Since there is not enough data to apply the same concept to electricity consumption, the idea of oversizing the electric capacity is explored to ensure that at all times, there is a certain amount of operating reserve (*OR*), i.e. additional electric capacity that can be brought online immediately, to capture unforeseen variations in electric load or generation from intermittent sources [11]. The concept of operating reserve can be extended to cover variations in hot water demand as well. The operating reserve is calculated as the difference between the actual generation at time t and the maximum amount of electricity that could be provided by the fuel cell (*FC*), the PV system (*PV*), and the battery (*BAT*) (eq. (3.16)).

$$\begin{aligned} OR_t = {} & \eta_{el}x_{FC,gas} - x^{-}_{FC,el,t} + x_{PV,el,t} - x^{-}_{PV,el,t} \\ & + \Delta t\left(\sum_{\tau=1}^{t}\epsilon^{t-\tau}\left(\eta^{+}y^{+}_{el,\tau} - \eta^{-}y^{-}_{el,\tau}\right)\right) - m_0 y_{BAT} \end{aligned} \tag{3.16}$$

In order to counter unforeseen changes in electricity consumption, the operating reserve is constrained to be always bigger than a fraction k of the difference between the peak load and the current load:

$$OR_t \geq k \cdot \left(\max_{t \in T}(L_{el,t}) - L_{el,t}\right) \tag{3.17}$$

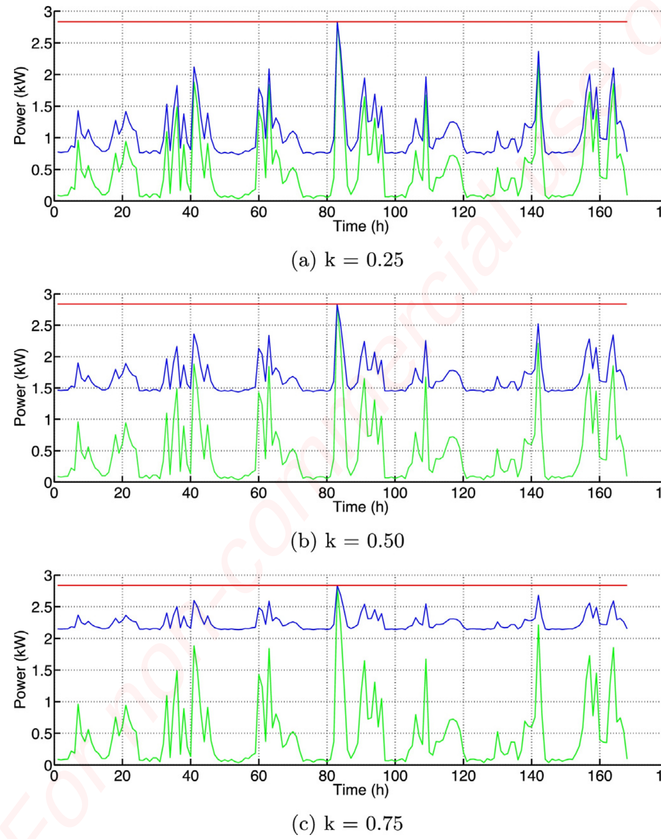


Fig. 6. Primary load (green) with operating reserve (blue) and maximum load (red) for different values of k. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

where $L_{el,t}$ only considers primary load. It does not include the electricity required to drive any of the conversion units such as the heat pump. Fig. 6 shows primary load (green) and primary load with operating reserve (blue) for one week and different values of k. As long as the actual electric load is below the blue line, the system will be able to cover it. Thus the higher k, the more risk-averse the optimization procedure, as it considers a larger amount of operating reserve.

4. Results and discussion

In a first part it is shown how ORES (Optimization of Residential Energy Systems) performs as compared to a commercially available tool for microgrid design (HOMER). A second part investigates how the use of alternative energy systems can reduce yearly energy costs and carbon dioxide emissions as compared to a reference case where all electricity is imported from the grid and all heat is provided by a gas boiler.

The test case for the comparison between ORES and HOMER is to satisfy the electricity needs of an off-grid house. The conversion and storage units under consideration are a cogeneration fuel cell, PV panels and a battery. HOMER uses net present cost as a metric to evaluate the life-cycle cost of the energy system. In order to allow a comparison with the annualized costs used in ORES, a project lifetime of 25 years is assumed. The economic and technical parameters of the energy conversion and storage system are the same in both HOMER and ORES. Indeed, both optimizers consider a discount rate of 3% and a cost of 60 CHF per ton of CO_2. The fuel cell is modeled with a constant fuel efficiency curve, a lower operating threshold of 30% of its nominal gas flow rate, an electrical and thermal efficiency of 50 respectively 40% based on the lower

Table 2
Search space for HOMER.

PV size (kW)	Battery size (kWh)	FC size (kW_{el})
0.00	0.00	0.00
0.50	1.00	0.50
0.75	1.50	0.75
1.00	2.00	1.00
1.25	2.50	1.25
1.50	3.00	1.50
1.75	3.50	1.75
2.00	4.00	2.00
2.25	4.50	2.25
2.50	5.00	2.50
2.75	5.50	
3.00	6.00	
3.25	6.50	
3.50	7.00	
3.75	7.50	
4.00		
4.25		
4.50		
4.75		
5.00		

heating value of natural gas, no thermal cycles, a very optimistic installation cost of 4000 CHF/kW_{el}, a maintenance cost of 200 CHF per year, and a lifetime of 15 years. The cost of natural gas is assumed to be 13.18 ct/kWh_{LHV}. For the battery, HOMER does not model self-discharge, which is why the round-trip efficiency for the HOMER battery is reduced from 92 to 90%. Otherwise all parameters such as maximum charge and discharge power and costs are kept the same in both ORES and HOMER. Finally, the PV system is modeled with an efficiency of 16% and a performance ratio of 90%. HOMER requires the modeling of a converter, which is not included in ORES. An ideal converter is thus included into the HOMER model, where ideal means that it is lossless and does not affect the system performance or the complexity of the optimization. While this work was under review HOMER released a new genetic algorithm inspired optimization module which eliminates the need to specify discrete design points for the battery and the PV system, but not for the fuel cell. In the following ORES will be benchmarked against both algorithms. For an electric demand of about 3500 kWh based on simulations by Richardson [39] and solar radiation data based on Meteotest, ORES achieved about 10% savings in yearly energy cost and a 23% reduction in yearly carbon emissions as compared to the solution proposed by HOMER. Furthermore, ORES was 10 times faster in finding the solution than the traditional HOMER approach. The new HOMER genetic algorithm inspired optimization module only takes about three times longer than ORES. The solutions it proposes are almost the same in terms of cost than for the traditional version. However, they result in 7% higher CO_2 emissions. Neither model considers uncertainty in the optimization process.

Since HOMER is based on a simulation approach, a variety of candidate solutions needs to be declared (Table 2). Through the use of decision rules, the performance of each solution is evaluated, which in turn allows to rank the candidate solutions. The optimization time depends on the number of candidate solutions. The optimal solutions proposed by HOMER and ORES are detailed in Table 3.

The battery is used quite differently in ORES and HOMER. HOMER uses decision rules to determine the battery – fuel cell dispatch. Two strategies are implemented: cycle-charging and load-following. Cycle-charging means that whenever the fuel cell is active, HOMER will try to run it at rated capacity and charge the battery with the excess electricity. Load-following means that the fuel cell will never be used to charge the battery. The results in Table 3 are obtained for the cycle-charging algorithm. If load-following is applied, this increases both yearly energy costs and CO_2 emissions. ORES has perfect knowledge of the electricity demand and solar radiation over an entire year. It uses the battery more effectively to match electricity demand and production. This results in a better utilization factor of the energy conversion units, which finally translates into savings on the investment cost. The HOMER solution has an excess electricity production of about 1800 kWh. This is because HOMER measures the electricity that could be produced by the installed PV system and not the PV electricity that is actually used in the system. Similarly, for the fuel cell production, HOMER indicates the amount of electricity that was produced by the fuel cell – which since the cell is constrained to run at least at 30% of its rated capacity – is higher than the power that is required to meet demand. Opposed to that, ORES measures how much energy each unit contributed to meet the electric load. It furthermore appears, that HOMER does sometimes during winter incur capacity shortages. At peak times, there is a 520 W capacity missing. These situations arise quite rarely. Over an entire year, there are 2.8 kWh or less than one percent missing. Since ORES is based on mathematical programming and since no uncertainty is considered, it is always able to meet demand. Overall, the solutions found by the algorithms are in the same range. This gives a first validation of ORES – since both programs are solving the same problem, it is expected that they yield similar results.

Five scenarios are analyzed to gage the impact of using ORES to plan and operate residential energy systems:

Business as usual	corresponds to a household that covers its electricity needs with imports from the grid, while heat needs are satisfied by a boiler.
Offgrid	corresponds to a home that does not exchange electricity with the grid. It does however have access to natural gas (which may be stored in tanks).
No exports	corresponds to a home that is connected to both the electricity and the gas grid. All technologies can be used, but produced electricity can not be exported to the grid.

Table 3
Comparison between HOMER and ORES results.

	Costs (CHF yr^{-1})	CO_2-emissions (kg yr^{-1})	PV		Battery		Fuel cell	
HOMER: 6600 simulations in 5:44 min								
Size	2639	1633	1000	W	4.00	kWh	1000	W
Throughput			1204	kWh	341	kWh	4109	kWh
HOMER with optimizer: 989 simulations in 1:36 min								
Size	2653	1752	240	W	4.50	kWh	1000	W
Throughput			289	kWh	435	kWh	4408	kWh
ORES: Model time – 24.1 s; Solver time – 5.7 s								
Size	2353	1263	765	W	4.47	kWh	960	W
Useful throughput			674	kWh	1174	kWh	2971	kWh

Table 4
The parameters describing the conversion technologies.

Parameter	Name	Unit	Value	Reference
Cogeneration fuel cell				
Targeted installation cost	c_{INV}	CHF kW_{el}^{-1}	4000	Estimate
Maintenance cost	c_{OMEX}	CHF kW_{el}^{-1} yr^{-1}	200	Estimate
Fuel cost	c_{fuel}	cts kWh_{gas}^{-1}	13.18	Services Industriels de Lausanne [20]
Lifetime	T_{life}	yr	15	Estimate
Annualized investment and maintenance cost	c_{cap}	CHF kW_{el}^{-1}	535	Calculation
Electric efficiency	η_{el}	%	50	SolidPower [25]
Thermal efficiency	η_{th}	%	40	SolidPower [25]
CO_2 emissions	ρ_{CO_2}	g kWh_{gas}^{-1}	200	Calculated
Minimum gas flow	k	% of nominal rating	30	SolidPower [25]
Photovoltaic system				
Installation cost	c_{INV}	CHF kW_{el}^{-1}	2500	Estimate based on ISE [28]
Maintenance cost	c_{OMEX}	CHF kW_{el}^{-1} yr^{-1}	36	Swiss Energyscope [31]
Lifetime	T_{life}	yr	25	Swiss Energyscope [31]
Annualized investment and maintenance cost	c_{cap}	CHF kW_{el}^{-1}	180	Calculation
Electric efficiency	η_{el}	%	16	ISE [28]
Performance ratio	η_{pr}	%	90	ISE [28]
Power density	ρ	W_{el} m^{-2}	144	Calculation based on standard rating conditions
Solar thermal system				
Installation cost	c_{INV}	CHF kW_{th}^{-1}	770	Swiss Energyscope [31], IEA [46]
Maintenance cost	c_{OMEX}	CHF kW_{th}^{-1} yr^{-1}	28	Swiss Energyscope [31]
Lifetime	T_{life}	yr	20	Swiss Energyscope [31]
Annualized investment and maintenance cost	c_{cap}	CHF kW_{th}^{-1}	80	Calculation
Power density	ρ	W_{th} m^{-2}	650	German Society for Solar Energy Systems [47]
Absorption efficiency	η	%	80	German Society for Solar Energy [47]
Heat losses	k	$W m^{-2} K^{-1}$	4	German Society for Solar Energy [47]
Heat pump				
Installation cost	c_{INV}	CHF kW_{th}^{-1}	1500	Swiss Energyscope [31]
Maintenance cost	c_{OMEX}	CHF kW_{th}^{-1} yr^{-1}	99	Swiss Energyscope [31]
Lifetime	T_{life}	yr	23	Swiss Energyscope [31]
Annualized investment and maintenance cost	c_{cap}	CHF kW_{th}^{-1}	190	Calculation
Coefficient of performance	η_{COP}	kW_{th} kW_{el}^{-1}	4	Calculation
Fuel cost	c_{fuel}	cts kWh_{th}^{-1}	5.55	Calculation
CO_2 emissions	ρ_{CO_2}	g kWh_{el}^{-1}	91.5	Swiss Federal Office of Energy [1]
Condensing gas boiler				
Installation cost	c_{INV}	CHF kW_{th}^{-1}	178	Swiss Energyscope [31]
Maintenance cost	c_{OMEX}	CHF kW_{th}^{-1} yr^{-1}	14	Swiss Energyscope [31]
Fuel cost	c_{fuel}	cts kWh_{gas}^{-1}	11.86	Services Industriels de Lausanne [20]
Lifetime	T_{life}	yr	17	Swiss Energyscope [31]
Annualized investment and maintenance cost	c_{cap}	CHF kW_{el}^{-1}	28	Calculation
Thermal efficiency	η	%	95	Based on Swiss Energyscope [31]
CO_2 emissions	ρ_{CO_2}	g kWh_{gas}^{-1}	179	Calculated

Net-metering represents the case, where electricity exports can amount up to the total yearly consumption. In such a case, the household only pays the net electricity consumed during a given year.

Feed-in-tariffs allow for unlimited electricity exports that are retributed at the same price as electricity imports.

In the business as usual case, ORES optimizes the control but not the investment planning of the energy system. In all other cases, it optimizes both investment and control. In all ORES-designed systems, the CO_2 emissions are lower than in the business as usual case (Fig. 7a). Feed-in-tariffs allow to reduce emissions by about 90%. Unsurprisingly, the yearly energy costs are higher for the offgrid case than for the grid-connected scenarios. In fact, being independent of the electricity grid increases the yearly energy bill by 30% as compared to business as usual. The cogeneration fuel cell is used in combination with a battery and a hot water tank to meet the variable electric and thermal loads. In the absence of any policies that encourage the use of renewables, i.e. in the no exports case, yearly energy costs drop by 5%, emissions however are curved by 45%. The reduction is achieved through the use of a heat pump in combination with a small solar thermal and a PV installation. The increase in electricity consumption due to the heat pump is more than offset by the savings in the natural gas consumption. This result relies on the low carbon intensity of the Swiss electricity mix, which consists mostly of hydro and nuclear power. The results may be different in countries with more coal and gas based electricity production. The carbon intensity of the German electricity mix for example is six times higher than the Swiss one (600 instead of 92 gCO_2 eq per kWh [44,45]). Finally, feed-in-tariffs favor mostly the installation of solar power (Fig. 7b). The PV electricity exports reduce the yearly energy bill by 60%.

It has been shown that ORES yields slightly better but similar results than the microgrid simulation software HOMER. The main difference lies in ORES' shorter computation time, the fact that it does not require the definition of a discrete search space, and its predictive dispatch algorithm. Due to its simulation based nature, HOMER allows to capture the physics of the energy system in more detail. Most prominently, it allows to model non-linearities. ORES has an advantage in calculating the system dispatch, since it considers load and production forecasts, whereas HOMER implements fixed strategies that do not make use of forecasts. The difference in the system dispatch is most prominent in the use of the battery. Both tools rely on load and generation estimates for the investment decisions. However, since HOMER calculates the

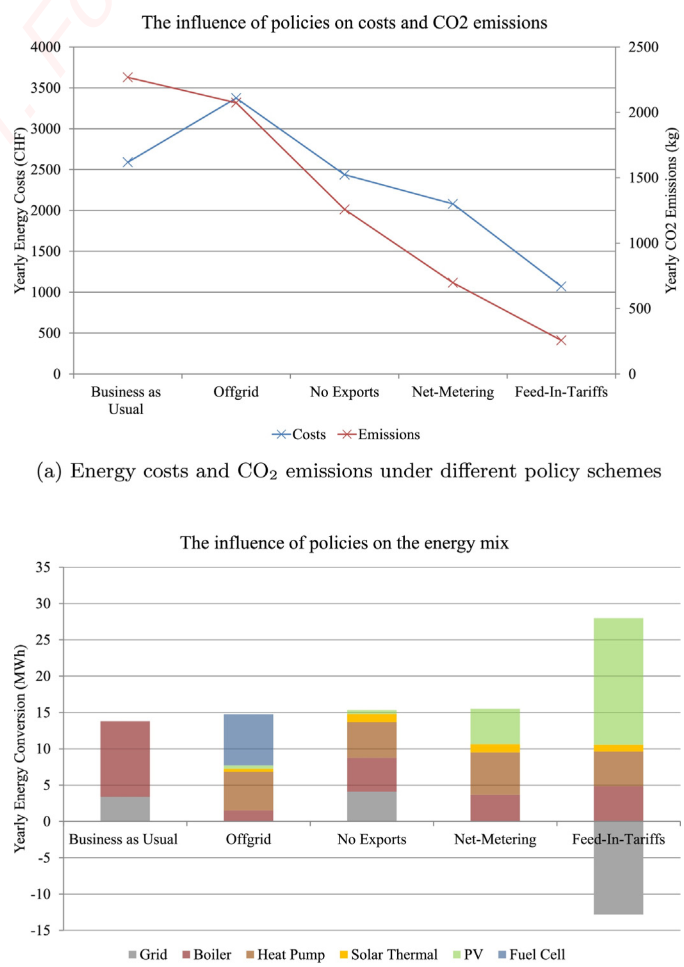
Table 5
The parameters describing the storage technologies.

Parameter	Name	Unit	Value	Reference
Battery				
Installation cost	c_{INV}	CHF kWh_{el}^{-1}	1000	Victron energy [33]
Lifetime	T_{life}	yr	9	Victron energy [33]
Annualized investment and maintenance cost	c_{cap}	CHF kWh_{el}^{-1}	128	Calculation
Roundtrip efficiency	η	%	92	Victron energy [33]
Self-discharge	ϵ	% $24\,h^{-1}$	5	Estimate
Max. depth of discharge	$1-m_0$	%	80	Victron energy [33]
Max. charging power	ρ^{+}	W Wh^{-1}	0.58	Victron energy [33]
Max. discharging power	ρ^{-}	W Wh^{-1}	1.00	Victron energy [33]
Initial state of charge	y_0	%	100	Assumption
Terminal state of charge	y_T	%	100	Assumption
Heat store				
Installation cost	c_{INV}	CHF kWh_{th}^{-1}	86	Estimate
Lifetime	T_{life}	yr	20	Swiss Energyscope [31]
Annualized investment and maintenance cost	c_{cap}	CHF kWh_{th}^{-1}	6	Calculation
Diameter	ϕ	cm	80	Assumption
Heat loss coefficient	U	W $m^{-2}K^{-1}$	0.21	German Society for Solar Energy [47]
Warm water outlet temperature	T_{hot}	°C	60	Legal requirement [26]
Water inlet temperature	T_{cold}	°C	10	Stadtwerke Karlsruhe [35]
Specific heat of water	c_p	J $kg^{-1}K^{-1}$	4.19	
Density of water	ρ	$kg\,l^{-1}$	1	
Average building temperature	T_{bdg}	°C	23	Assumption
Initial state of charge	y_0	%	50	Assumption
Terminal state of charge	y_T	%	50	Assumption

system dispatch based on decision rules, it is expected to be more robust towards uncertainty. The ORES dispatch shows the value of perfect information. Finally, because of its continuous nature, ORES is guaranteed to find the optimal investment decisions – but it may not be possible to implement these solutions. HOMER, on the contrary, evaluates user defined scenarios. A way to combine the

Table 6
The decision variables used to model the conversion and storage technologies.

Decision variable	Unit	Description
Cogeneration fuel cell		
x_{gas}	W	Fuel cell chemical capacity
$x^{+}_{gas,t}$	W	Gas flow during time-step t
$x^{-}_{sph,t}$	W	Space heat provided during time-step t
$x^{-}_{dhw,t}$	W	Hot water provided during time-step t
$x^{-}_{el,t}$	W	Electricity provided during time-step t
Photovoltaic system		
x_{el}	W	PV capacity
$x^{-}_{el,t}$	W	Electricity provided during time-step t
Solar thermal system		
x_{th}	W	Solar thermal capacity
$x^{-}_{sph,t}$	W	Space heat provided during time-step t
$x^{-}_{dhw,t}$	W	Domestic hot water provided during time-step t
Heat pump		
x_{el}	W	Heat pump capacity
$x^{-}_{sph,t}$	W	Space heat provided during time-step t
Condensing gas boiler		
x_{gas}	W	Boiler capacity
$x^{+}_{gas,t}$	W	Gas flow during time-step t
$x^{-}_{sph,t}$	W	Space heat provided during time-step t
$x^{-}_{dhw,t}$	W	Domestic hot water provided during time-step t
Battery		
y_{el}	J	Battery capacity
$y^{+}_{el,t}$	W	Charging power during time-step t
$y^{-}_{el,t}$	W	Discharging power during time-step t
Heat store		
y_h	m	Height of the heat store
$y^{\pm}_{dhw,t}$	W	Power inflow (positive) or outflow (negative) during time-step t



(a) Energy costs and CO_2 emissions under different policy schemes

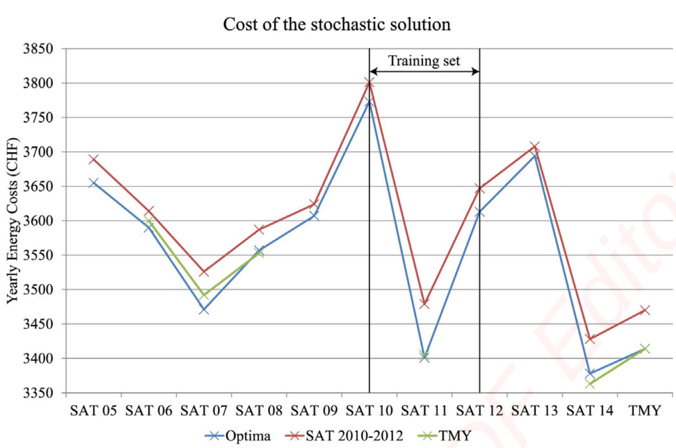
(b) The composition of the energy mix under different policy schemes

Fig. 7. The influence of policies on residential energy consumption.

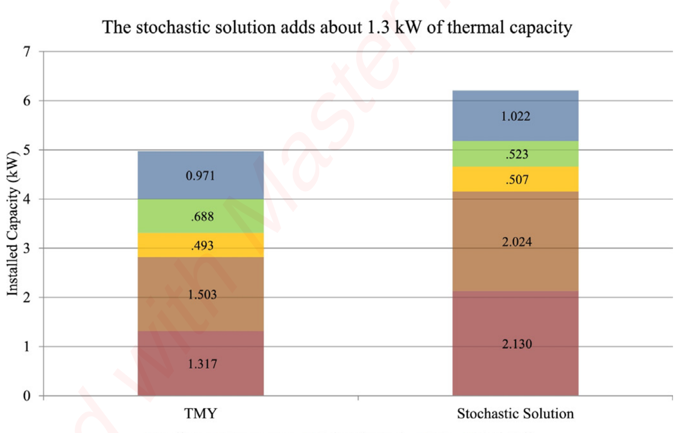
advantages of both approaches is to use mathematical programming to find promising candidate solutions, which are evaluated through a more advanced simulation in a second stage.

It has been demonstrated that ORES can be used to analyze the impact of policies on residential energy systems. Combining today's available technologies, it is possible to reduce yearly energy costs and CO_2 emissions even if there are no additional incentives that encourage the use of renewable energy. It is interesting to note that while an optimized system results in only 5% savings on the yearly energy bill, emissions are cut by up to 45%. These results are obtained without considering uncertainty in energy load and production.

When considering uncertainty in weather parameters, it appears that the stochastic solution based on the years 2010–2012 is feasible in all weather scenarios and yields only slightly higher yearly energy costs than the optimal solution for each year, where the investment decisions are not held constant but calculated for each year separately (Fig. 8a). The difference between the two, referred to as the *Expected Value of Perfect Information* (EVPI) is just 1.2%. The cost of the solution (in green) based on a typical meteorological year is only plotted for years in which it does not violate comfort constraints – except for the year 2014, where there is only a minor shortage of electricity. As



(a) The stochastic solution is feasible in all cases and yields only slightly higher costs than the optimal solution for each scenario



(b) The installed energy conversion capacity for the TMY and the stochastic solution

Fig. 8. The impact of using the stochastic solution on energy costs and installed capacity.

expected, it is lower than the stochastic solution. The electric generation capacity for off-grid scenarios does not vary much between the stochastic and the typical meteorological year solution (Fig. 8b). The thermal capacity however, consisting of the investments in the boiler and the heat pump, increased by 47%. This result suggests that temperature variations which influence heat demand have a greater impact on the system configuration than solar radiation variations which mostly influence the photovoltaic production. Yearly energy costs rise only slightly because most of the capacity increase is covered by installing a larger boiler, which is the unit with the lowest investment cost.

Overall, it appears that stochastic programming proves to be a valuable tool in determining investment decisions that are robust towards uncertainty in weather parameters.

The main impact of imposing an operating reserve is an increase in the installed battery capacity. If no operating reserve is required, a 4.5 kWh battery is selected. This figure rises to 6.6 kWh, i.e. by almost half, if it is assumed that peak loads can occur at all timesteps ($k = 1$) and triggers a 9% increase in yearly energy costs. The reason for the battery's capacity increase by almost 50% is that now the system needs to be prepared to cover unforeseen electric peak loads at all time, which can be achieved through an increase in dispatchable generation (which in this case is set by the fuel cell capacity), an increase in storage capacity (given by the battery size), or a combination of the two. Indeed, ORES increases the fuel cell slightly from 1 to 1.018 kW and decreases the amount of (intermittent) photovoltaic capacity from 567 to 448 W. The most drastic change lies in the battery capacity, which is due to two reasons. First, the fuel cell and the PV system generate enough electricity to maintain a high state of charge of the battery to cover unforeseen load variations. Second, the cost of expanding battery capacity is about eight times lower than the cost of expanding fuel cell capacity.

5. Conclusion

A modular, flexible linear program (ORES) that determines the design and operating decisions for energy conversion and storage units in residential dwellings is developed. The optimization problem allows to study the impact of different policies such as feed-in tariffs, net-metering, and carbon-taxation. Furthermore, the influence of changes in weather, energy demand and technology parameters such as conversion efficiencies, investment and fuel costs can be analyzed.

A comparison with the commercially available microgrid design software package HOMER shows that ORES achieves similar but slightly lower energy costs in significantly less time, although this comparison arguably depends on the size and granularity of the search space considered by HOMER. Furthermore, ORES covers thermal loads and conversion units in more detail than HOMER. A first examination of the utility of combining several conversion and storage units shows that energy costs can be reduced by 5 to 60 percent, whereas CO_2 emissions drop by 45–90% depending on how the dwelling is allowed to exchange electricity with the grid. These results are obtained without putting a price on carbon and are compared to a reference case where all heat is provided by a natural gas boiler and all electricity is imported from the grid.

The model is made robust against weather and electric load uncertainty through stochastic programming and the consideration of operating reserve. It is wise to consider the stochastic solution, because the deterministic solution trained on a typical meteorological year is in half of the time not able to meet the residential energy demand. The Expected Value of Perfect Information is low which suggests that the stochastic solution performs well over all weather scenarios. On average the price of the stochastic solution is just 1.2% higher than the price of the deterministic solution. Uncertainty in the electric load is accounted for by decision rules based on the concept of operating reserve. In a conservative scenario, where it is assumed that the maximum electric load can occur at all times during the optimization horizon, the total energy costs rise by 9%. The concept can easily be extended to domestic hot water loads. During the literature research a lack of precise measurements on household energy consumption was observed. If more accurate data was available, this would allow the application of stochastic programming and robust optimization to uncertainty in energy consumption. Furthermore, chance-constraints could be used to model a budget of uncertainty corresponding to a certain risk of not satisfying energy demand that the building residents may be willing to take in exchange for lower energy bills.

Three directions are open for future work. First, the accuracy of the model can be improved by considering non-linearities in the conversion units, such as the fuel-efficiency curves for fuel cells operating in part mode. Second, one could try to relax the assumption that all parameters are known one year in advance and study the impact on yearly energy costs and CO_2 emissions. Third, a sensitivity analysis can be carried out to see under which conditions a certain technology becomes attractive from an economic or environmental point of view. This may be especially interesting for fuel cells, which are currently on the edge of commercialization.

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Appendix

Problem parameters

Decision variables

References

[1] Infras AG, Prognos AG, TEP Energy GmbH, Analyse des schweizerischen Energieverbrauchs 2000–2013 nach Verwendungszwecken, Tech. rep., Bundesamt für Energie, 2014 (30.09.2014).
[2] Simplified energy balances – annual data, Tech. rep., Eurostat, 2015, http://appsso.eurostat.ec.europa.eu/nui/submitViewTableAction.do.
[3] W. Buehring, C. Huber, J. Marques de Souza, Expansion Planning for Electrical Generating Systems: A Guidebook, Tech. Rep. 241, International Atomic Energy Agency, Vienna, 1984.
[4] S. Stoft, Power System Economics: Designing Markets for Electricity, IEEE Press, Wiley-Interscience, Piscataway, NJ and New York, 2002.
[5] A. Ashouri, Simultaneous Design and Control of Energy Systems (Ph.D. thesis), ETH Zürich, Zürich, 2014, http://e-collection.library.ethz.ch/view/eth:9057.
[6] A. Wright, S. Firth, The nature of domestic electricity-loads and effects of time averaging on statistics and on-site generation calculations, Appl. Energy 84 (4) (2007) 389–403, http://dx.doi.org/10.1016/j.apenergy.2006.09.008.
[7] C. Marnay, J.S. Chard, K.S. Hamachi, T. Lipman, M.M. Moezzi, B. Ouaglal, A.S. Siddiqui, A.A. Akhil, R.H. Lasseter, J.E. Dagle, J. Kueck, R.J. Yinger, Modeling of Customer Adoption of Distributed Energy Resources, Tech. rep., Lawrence Berkeley National Laboratory, 2001, http://escholarship.org/uc/item/6ts3c2pd.
[8] C. Marnay, G. Venkataramana, M. Stadler, A.S. Siddiqui, R. Firestone, B. Chandran, Optimal technology selection and operation of commercial-building microgrids, IEEE Trans. Power Syst. 23 (3) (2008) 975–982, http://escholarship.org/uc/item/3jj9r75m#page-1.
[9] C. Marnay, M. Stadler, A. Siddiqui, N. DeForest, J. Donadee, P. Bhattacharya, J. Lai, Applications of optimal building energy system selection and operation, Proc. Inst. Mech. Eng. A: J. Power Energy 227 (1) (2013) 82–93, http://dx.doi.org/10.1177/0957650912468408.
[10] M. Stadler, M. Groissböck, G. Cardoso, C. Marnay, Optimizing distributed energy resources and building retrofits with the strategic DER-CAModel, Appl. Energy 132 (2014) 557–567, http://dx.doi.org/10.1016/j.apenergy.2014.07.041.
[11] T. Lambert, P. Gilman, P. Lilienthal, Micropower System Modeling with Homer, IEEE Press, Wiley-Interscience, Piscataway and Hoboken, NJ, 2006.
[12] M.T. Iqbal, A feasibility study of a zero energy home in Newfoundland, Renew. Energy 29 (2) (2004) 277–289, http://dx.doi.org/10.1016/S0960-1481(03)00192-7.
[13] M. Beccali, S. Brunone, M. Cellura, V. Franzitta, Energy, economic and environmental analysis on RET-hydrogen systems in residential buildings, Renew. Energy 33 (3) (2008) 366–382, http://dx.doi.org/10.1016/j.renene.2007.03.013.
[14] S.M. Shaahid, M.A. Elhadidy, Economic analysis of hybrid photovoltaic-diesel-battery power systems for residential loads in hot regions – a step to clean future, Renew. Sustain. Energy Rev. 12 (2) (2008) 488–503, http://dx.doi.org/10.1016/j.rser.2006.07.013.
[15] T. Mahlia, P.L. Chan, Life cycle cost analysis of fuel cell based cogeneration system for residential application in Malaysia, Renew. Sustain. Energy Rev. 15 (1) (2011) 416–426, http://dx.doi.org/10.1016/j.rser.2010.07.041.
[16] Meteotest, Meteonorm: Irradiation Data for Every Place on Earth., 2005, http://meteonorm.com/en/.
[17] Entwicklung der Wechselrate im Strommarkt, Tech. rep., Eidgenössische Elektrizitätskommission ElCom, 2014 (14.11.14).
[18] V. Beurer, Neuordnung des Erdgassektors in der Schweiz, Tech. rep., Bundesamt für Energie, 2007 (23.04.07).
[19] Romande Energie, Simulateur de Facture, 2015, http://www.romande-energie.ch/particuliers/electricite-compte-et-factures/tarifs-et-simulateur/simulateur-facture.
[20] Services Industriels Lausanne, Tarifs du gaz chauffage, 2015, http://www.lausanne.ch/thematiques/services-industriels/particuliers/tarifs/tarifs-du-gaz.html.
[21] D. MacKay, Sustainable Energy – Without the Hot Air, UIT Cambridge, 2008 www.withouthotair.com.
[22] R. Braun, Optimal Design and Operation of Solid Oxide Fuel Cell Systems for Small-scale Stationary Applications (Ph.D. thesis), University of Wisconsin, Madison, 2002.
[23] J. Larminie, A. Dicks, Fuel Cell Systems Explained, 2nd ed., J. Wiley, Chichester, West Sussex, 2003.
[24] D. Carter, Latest Developments in the Ene-Farm Scheme, Tech. rep., FuelCellToday, 2013, http://www.fuelcelltoday.com/analysis/analyst-views/2013/13-02-27-latest-developments-in-the-ene-farm-scheme.
[25] SOLIDpower, EnGen-2500 Technical Specifications, 2014, http://www.solidpower.com/en/engen-2500/#engen2500.
[26] Legionella et eau potable – à quoi faut-il faire attention?, Tech. rep., Société Suisse de l'Industrie du Gaz et des Eaux, 2000.
[27] A.D. Hawkes, P. Aguiar, B. Croxford, M.A. Leach, C.S. Adjiman, N.P. Brandon, Solid oxide fuel cell micro combined heat and power system operating strategy: options for provision of residential space and water heating, J. Power Sources 164 (1) (2007) 260–271, http://dx.doi.org/10.1016/j.jpowsour.2006.10.083.
[28] H. Wirth, Aktuelle Fakten zur Photovoltaik in Deutschland, Tech. rep., Fraunhofer-Institut für Solare Energiesysteme ISE, 2015 (19.05.15).
[29] International Renewable Energy Agency, REsource, 2015, http://resourceirena.irena.org.
[30] Solar Rating And Certification Corporation, Standardized Comparisons of Solar Thermal Products, 2015, http://www.solar-rating.org/.
[31] Energy Center of EPFL, IPESE group, Swiss-Energyscope, 2015, http://calculator.energyscope.ch/.
[32] Battery Storage for Renewables: Market Status and Technology Outlook, Tech. rep., International Renewable Energy Agency, 2015.
[33] Victron energy, 12.8 Volt Lithium Iron Phosphate Batteries, 2015, https://www.victronenergy.com/batteries/lithium-battery-12-8v.
[34] Jenni Energietechnik, Preisliste, Tech. rep., 2015 www.jenni.ch.
[35] Stadtwerke Karlsruhe, Wasserqualität: Online-Werte Trinkwasser, 2015, http://www.stadtwerke-karlsruhe.de/swka-de/inhalte/produkte/trinkwasser/online-wert-trinkwasser.php.
[36] G.M. Masters, Renewable and Efficient Electric Power Systems, second ed., Wiley, 2013.
[37] J. Widén, M. Lundh, I. Vassileva, E. Dahlquist, K. Ellegård, E. Wäckelgård, Constructing load profiles for household electricity and hot water from time-use data – modelling approach and validation, Energy Build. 41 (7) (2009) 753–768, http://dx.doi.org/10.1016/j.enbuild.2009.02.013.
[38] U. Jordan, K. Vajen, Influence of the DHW load profile on the fractional energy savings, Sol. Energy 69 (2001) 197–208, http://dx.doi.org/10.1016/S0038-092X(00)00154-7.
[39] I. Richardson, M. Thomson, D. Infield, C. Clifford, Domestic electricity use: a high-resolution energy demand model, Energy Build. 42 (10) (2010) 1878–1887, http://dx.doi.org/10.1016/j.enbuild.2010.05.023.
[40] H. Park, M. Ruellan, A. Bouvet, E. Monmasson, R. Bennacer, Thermal Parameter Identification of Simplified Building Model With Electric Appliance, IEEE, Piscataway, NJ, 2011, pp. 1–6, http://dx.doi.org/10.1109/EPQU.2011.6128822.
[41] Monitoring Atmospheric Composition and Climate, Meteosat, Time Series of Solar Radiation Data from MACC-RAD, 2015, http://www.soda-pro.com/web-services/radiation/macc-rad.
[42] Goddard Earth Observing System, NASA, Time-Series of Surface Meteo Variables from MERRA Re-analysis, 2015, http://www.soda-pro.com/web-services/meteo-data/merra.

[43] A. Shapiro, D. Dentcheva, A.P. Ruszczynski, Lectures on Stochastic Programming: Modeling and Theory, Vol. 9 of MPS-SIAM Series on Optimization, Philadelphia, 2009.
[44] P. Stolz, R. Frischknecht, Umweltbilanz Strommix Schweiz, Tech. rep., Treeze Ltd., 2011.
[45] P. Icha, Entwicklung der spezifischen Kohlendioxidemissionen des deutschen Strommix in den Jahren 1990 bis 2012, Tech. rep., Umweltbundesamt, 2013.
[46] Solar Heating and Cooling for Residential Applications: Technology Brief, Tech. rep., IEA-ETSAP and IRENA, 2015.
[47] Deutsche Gesellschaft für Sonnenenergie, Planning and Installing Solar Thermal Systems: A Guide for Installers, Architects and Engineers, 2nd ed., Earthscan, London and Washington, DC, 2010.